Answer Key & Explanations

1. True (Friction always reduces output force, so AMA < IMA).
2. 3rd Class Lever (Effort is between fulcrum and load).
3. a) Increases (IMA = Length / Thickness; larger numerator = larger IMA).
4. Inclined Plane
5. False (A single fixed pulley has an IMA of 1; it only changes direction).
6. b) Work (Ideal machines: Work In = Work Out. Real machines: Work In > Work Out).
7. 3rd Class Lever (Effort applied by thumb in middle, hinge at back, staple at front). *Note: Some interpret staplers as 2nd class if pressing the top arm down on a desk, but "handheld" is usually 3rd.*
8. b) Less than 1 (You are applying force over a small distance to move the wheel a large distance—speed multiplier, force reducer).
9. 62.8 (Circumference = $\pi \times 40mm \approx 125.6mm$. Pitch = 2 mm. IMA = $125.6/2 = 62.8$).
10. 5 N (IMA = Radius Effort / Radius Load = $4/1=4$. Force = Load / IMA = $20/4=5$).
11. 4 (Count the supporting rope segments) & Pulley
12. 500 Joules ($W=F \times d=50 \times 10$).
13. 40 RPM (Ratio is 40:8=5:1. Speed decreases by factor of 5. $200/5=40$).
14. 12 (Total IMA = IMA1 × IMA2 for machines in series).
15. 250 N
    a. Load = 1000 N. IMA = 5. Ideal Force = 200 N.
    b. Efficiency = Ideal Force / Actual Force $\rightarrow 0.80=200/F_{actual} \rightarrow F_{actual} = 200/0.8 = 250$.

16. 72% ($0.90 \times 0.80 = 0.72$).

17. 3 meters
    a. Let x be distance from 5kg. Distance from 15kg is $(4-x)$.
    b. $5x = 15(4-x) \Rightarrow 5x = 60 - 15x \Rightarrow 20x = 60 \Rightarrow x = 3$.

18. 200 N
    a. Ideal Force = Load / IMA = $2000/5 = 400$ N.
    b. Actual Force = 600 N.
    c. Lost Force = $600 - 400 = 200$ N.

19. Wedge (or Inclined Plane friction analysis). It relies on the angle of repose/friction coefficient $\tan(\theta) \leq \mu$.

20. Archimedes

21. B (Total IMA = 3 * 2 = 6. AMA = IMA * Efficiency = 6 * 0.80 = 4.8)

22. D (IMA of screw = Circumference of handle / Pitch. Decreasing circumference reduces the numerator, lowering IMA.)

23. C (Brooms are Class 3: One hand is the fulcrum at the top, the other applies effort in the middle, load is at the bottom. They increase speed/distance, not force.)

24. C (Static equilibrium usually implies rest, but "Dynamic Equilibrium" (D) means moving at constant velocity. However, in the context of Static equilibrium specifically, the object must be at rest. But typically in physics tests, the definition of equilibrium (net force/torque = 0) applies to both. The "trick" here is that equilibrium allows for constant velocity (D), so D is a true statement about equilibrium mechanics generally, but the term "Static" implies rest. If we treat "Static Equilibrium" strictly, C is true. If we look for the FALSE statement regarding the conditions for equilibrium ($F_{net}=0$, $t_{net}=0$), D is the scenario that is *not* "static". *Self-Correction for clarity:* Let's look at standard test definitions. "Static equilibrium" strictly means the object is at rest. "Dynamic equilibrium" means constant velocity. The question asks what is FALSE. A, B are conditions for *any* equilibrium. D is describing *dynamic* equilibrium. Therefore, D is the False statement *in the context of defining Static Equilibrium*. Wait, actually C says "The

object MUST be stationary". This is TRUE for *Static* Equilibrium. D says "The object can be moving". This is FALSE for *Static* Equilibrium. So D is the correct answer (it is the false statement).)

25. C (Work is Force * Distance. The ramp reduces force but increases distance. In a frictionless ideal world, Work In = Work Out (mgh).)

26. C (Joule = Newton-meter. N . s is Impulse. kg . m/s^2 is Newton. W/s is meaningless/Change in power.)

27. A (IMA of wedge = Length / Thickness. If thickness doubles (2 to 4), the denominator doubles, so the IMA is halved.)

28. C (IMA = Effort Arm / Load Arm. To maximize IMA, you want the Effort Arm to be as long as possible and the Load Arm to be as short as possible. Placing the fulcrum close to the load achieves this.)

29. B (A screw is conceptually an inclined plane wrapped around a central shaft.)

30. B (AMA = Output Force / Input Force. If AMA = 1, then Output Force = Input Force.)

31. D (Torque = F . d . sin(theta). sin(90) is 1 (maximum). sin(0) is 0.)

32. B (Power = Work / Time.)

33. B (IMA of ramp = Length / Height. Doubling length doubles IMA. Force = Load / IMA. If IMA doubles, Force is halved.)

34. C (A single fixed pulley has an IMA of 1. It offers no mechanical advantage in force, only changes direction.)

35. B (Work requires displacement. Pushing a wall that doesn't move means distance = 0, so Work = 0.)

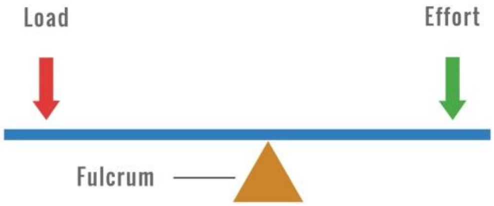
36. B (Efficiency = Output / Input = 40 / 50 = 0.8 or 80%.)

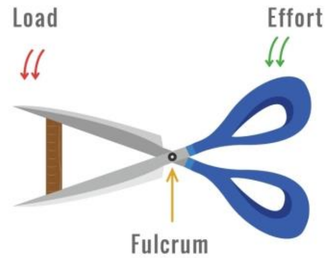
37. A (Speed Ratio is inverse to tooth count. 10 teeth/ 50 teeth = 1/5. 100 RPM / 5 = 20 RPM)

The Three Lever Classes

Types of Levers | Examples

First Class





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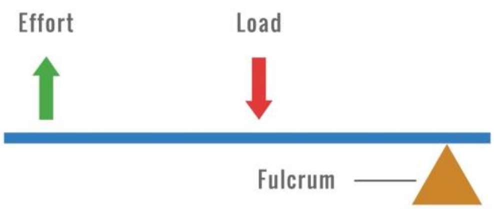

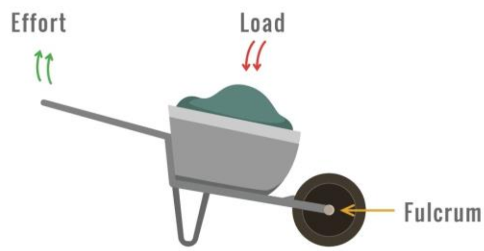

Third Class